

Ex 6: Programs in Prolog

1. Family Relationships

Problem: Define family relations and answer grandparent queries.

% Facts

parent(john, mary).

parent(mary, susan).

parent(john, mike).

% Rules

grandparent(X, Y) :- parent(X, Z), parent(Z, Y).

% Query

% ?- grandparent(john, susan).

Output:

16 ?- consult(['c:/Users/HDCO422166/Downloads/family.pl']).

true.

17 ?- grandparent(john, susan).

true .

2: Animal Classification

Problem: Identify if an animal is a mammal based on features.

% Facts

has_hair(tiger).

gives_milk(tiger).

% Rule

mammal(X) :- has_hair(X), gives_milk(X).

% Query

% ?- mammal(tiger).

Output:

18 ?- consult(['c:/Users/HDCO422166/Downloads/animal.pl']).

true.

19 ?- mammal(tiger).

true.

3: Simple Arithmetic Knowledge Base

Problem: Implement simple addition using PROLOG.

% Rule

add(X, Y, Z) :- Z is X + Y.

% Query

% ?- add(5, 7, Z).

Output:

20 ?- consult(['c:/Users/HDCO422166/Downloads/math.pl']).

true.

21 ?- add(5, 7, Z).

Z = 12.